Lab: Level Up -- Tracking How You Learn

Objective

> **Learning Goal:** Investigate how learning works by tracking your own skill improvement, understanding model updating, and discovering what makes practice effective.

This is a **group challenge** designed for middle school students (Grades 6-8).

Materials Needed

- Pencil and paper
- Timer or stopwatch
- A simple skill to practice (paper airplane folding, coin stacking, or word puzzle)
- Lab journal or notebook

Part 1: The Learning Curve (15 min)

You are going to learn something new (or improve at something) and track your progress in real time.

Choose a task: Paper airplane distance throw, stacking coins into a tower, solving simple anagram puzzles, or drawing a star with your non-dominant hand.

Do the task 6 times. After each attempt, record your result.

Attempt	Score/ Result	What did you change from last time?	Prediction error (expected vs. actual)
1		Nothing (baseline)	
2			
3			
4			
5			
6			

Write your response here

Part 2: What Changed in Your Brain? (10 min)

When you learn, your brain updates its internal model. Think about what changed between your first and last attempt.

Before Learning	After Learning
What did you expect would happen?	What do you expect now?
What strategy did you use?	What strategy do you use now?
How confident were you (1-10)?	How confident are you now (1-10)?
What surprised you?	What no longer surprises you?

Write your response here

Part 3: Learning Strategies Audit (10 min)

Not all learning strategies are equal. Rate how well each strategy works for YOU on a scale of 1-5.

Strategy	How often you use it (1-5)	How well it works (1-5)	Active Inference connection
Re-reading notes			Low prediction error -- brain is not challenged
Practice testing			High prediction error -- brain must update
Teaching someone else			Forces you to build a complete model
Watching videos passively			Low engagement, minimal updating
Making mistakes and correcting them			Direct prediction error drives learning
Spacing out practice over days			Lets the brain consolidate model updates

Write your response here

Part 4: Design Your Study Plan (5 min)

Pick something you are currently trying to learn (a school subject, a hobby, a skill). Design a study plan based on what you now know about how learning works.

Element	Your Plan
What are you learning?	
What is your current mental model (what do you know so far)?	
What are the biggest gaps (highest uncertainty)?	
What actions will generate useful prediction errors?	
How will you know your model has updated?	

Write your response here

Reflection Table

Question	Your Answer
What is a learning curve?	
How does your brain's model change when you learn?	
Why are mistakes important for learning?	
What is the difference between feeling like you learned and actually learning?	
How does Active Inference explain learning?	

Write your response here